

Hashing

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Hashing

What is Hashing?

Hashing is a fundamental concept in computer science. It's the process of converting an input of any length (like a string of text) into a fixed-size, unique output. This output is typically a number and is called a "**hash code**" or simply a "**hash**".

The magic behind hashing is performed by a **hash function**. A good hash function always produces the same hash for the same input and is designed to produce different hashes for different inputs.

The most common application of hashing is in **hash tables** (also known as hash maps). A hash table is a data structure that allows for incredibly fast data retrieval. Instead of searching through an entire list to find an item, you can use its hash to find its location almost instantly.

Types:

1. Separate Chaining
2. Linear Probing
3. Quadratic Probing

Hashing

apple, orange, grape, melon, plum, kiwi, lemon, peach

10, 20, 30, 100, 110, 120, 130, 140

1. Separate Chaining

Hash Function Formula:

$$\text{index} = (\text{SUM of ASCII values}) \% 10$$

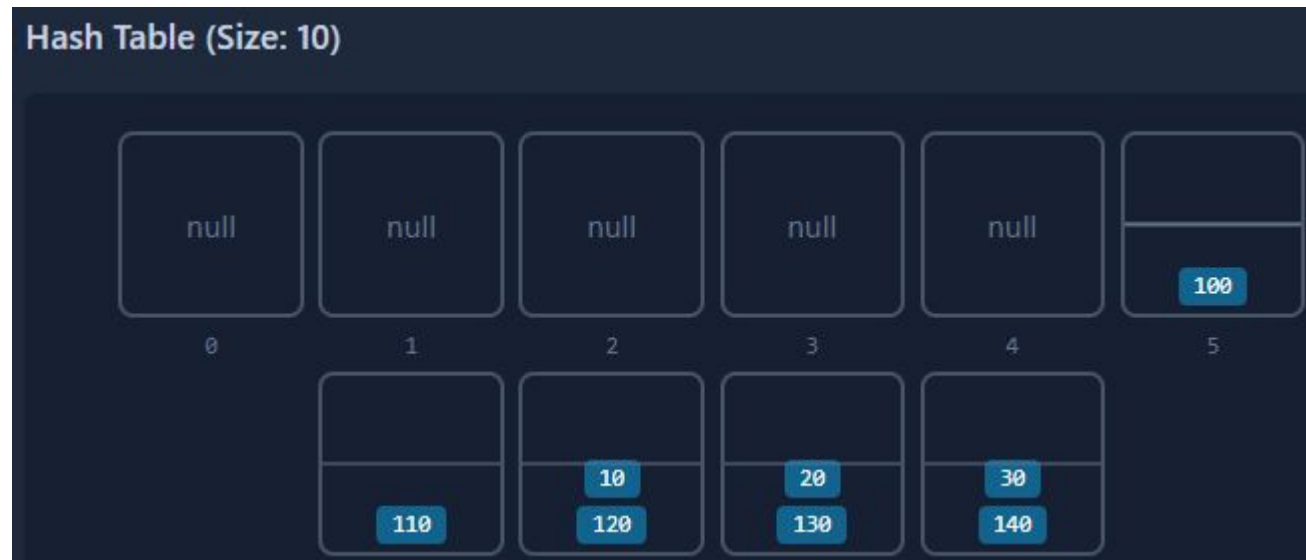
Hash Calculation

Sum of ASCII values for "apple":

$$97 + 112 + 112 + 108 + 101 = \mathbf{530}$$

Hashing key "apple". Sum = 530. Hash = $530 \% 10 = 0$.

Same Hash Function



Hashing

1. Linear Probing

Hash Function Formula:

$$\text{index} = (\text{SUM of ASCII values}) \% 10$$

Hash Calculation

Sum of ASCII values for "apple":

$$97 + 112 + 112 + 108 + 101 = \mathbf{530}$$

Hashing key "apple". Sum = 530. Hash = $530 \% 10 = 0$.

Check the next cell
if it is occupied

Same Hash Function



Hash Table (Size: 10)

apple	kiwi	lemon	peach	null	null
0	1	2	3	4	5
	orange	grape	plum	melon	
	6	7	8	9	

Hash Table (Size: 10)

120	130	140	null	null	100
0	1	2	3	4	5
	110	10	20	30	
	6	7	8	9	

Hashing

1. Quadratic Probing

```
"apple" = 530  
"orange" = 636  
"grape" = 527  
"melon" = 539  
"plum" = 446  
"kiwi" = 436  
"lemon" = 539  
"peach" = 513
```

For numeric strings:

```
"10" = 97  
"20" = 98  
"30" = 99  
"100" = 145  
"110" = 146  
"120" = 147  
"130" = 148  
"140" = 149
```

Formula:

$(\text{SUM of ASCII values}) \% 10$

For "apple":

$5 + 112 + 108 + 101 = 530$

Sum = 530. Hash = $530 \% 10 = 0$.

Same Hash Function



Thank you